

Recurrence and Ergodic Theorems

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Notation

Term	Description
$\mathbb{N}$	The Natural Numbers: 1, 2, 3, ...

Theorem 0.1 (Bergelson, 1985, Theorem 1.1). *Let $(X, \mathcal{B}, \mu)$ be a probability space and suppose that $B_n \in \mathcal{B}$ such that $\mu(B_n) = b > 0$ for all $n \in \mathbb{N}$.*

Then there exists a positively dense index set $I \subset \mathbb{N}$ such that, for any finite subset $F \subseteq I$, we have

$$\mu\left(\bigcap_{i \in F} B_i\right) > 0.$$

Theorem 0.2 (Poincaré Recurrence for Upper Density). *Let M be a countably-infinite amenable group with Φ as a Følner sequence. For all $B \in \mathcal{P}(\Gamma)$ such that $d_\Phi(B) > 0$, there exists $\gamma \in \Gamma$ such that $\bar{d}_\Phi(B \cap \gamma \cdot B) > 0$.*

This can be proved by using Furstenberg's Correspondence Principle with B as our subset and, as μ is measure-preserving in our Furstenberg system and by using [the invariance of density](#), we can show that

$$d_\Phi(B) = \mu(E) = \mu(\gamma \cdot E) = d_\Phi((\gamma) \cdot B)$$

for all $\gamma \in \Gamma$. By defining an infinite series $E_n = g_n \cdot E$ and knowing that $\mu(E_n) > 0$ for all $n \in \mathbb{N}$, we can use [Theorem 0.1](#) to arrive at the result that $\mu(h \cdot B \cap h' \cdot B) > 0$ for some $h, h' \in \Gamma$. By using the fact that μ is measure-preserving again and using the properties of μ as a result of its construction in Furstenberg's Correspondence Principle, we conclude that $\bar{d}_\Phi((\gamma) \cdot B \cap g \cdot B) > 0$ for some $\gamma \in \Gamma$

Theorem 0.3 (cf. Lindenstrauss, 2001, Theorem 1.2). *Let Γ be a discrete amenable group acting on a measure space $(X, \mathcal{B}, \mu)$ by measure preserving transformation and let $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ be a tempered Følner sequence.*

Then, for any $f \in L^1(\mu)$, there is a Γ -invariant $\bar{f} \in L^1(\mu)$ such that

$$\lim_{N \rightarrow \infty} \frac{1}{\lambda \Phi_N} \sum_{\gamma \in \Phi_N} f(\gamma.x) = \bar{f}(x)$$

for μ -almost every $x \in X$. In particular, if the Γ action is ergodic, then

$$\lim_{N \rightarrow \infty} \frac{1}{\lambda \Phi_N} \sum_{\gamma \in \Phi_N} f(\gamma.x) = \int f(x) d\mu(x)$$

for μ almost every x .

Corollary 0.1 (cf. Host, 2019, Corollary 8). *Let (X, Γ) be a topological dynamical system where Γ is an amenable group, μ an ergodic measure on X and Φ a tempered Følner sequence. Then μ -almost every $x \in X$ is generic for μ along Φ .*

Theorem 0.4 (Jamneshan and Kreidler, 2025, Abstract Mean Ergodic Theorem). *For every contraction semi-group $\mathcal{S} \subseteq \mathcal{L}(H)$ on a Hilbert space H , we have an orthogonal decomposition*

$$H = \text{fix}(\mathcal{S}) \oplus \overline{\text{lin}} \bigcup_{U \in \mathcal{S}} (Id_H - U)(H).$$

Moreover, the orthogonal projection P onto $\text{fix}(\mathcal{S})$ has the following properties:

1. $PU = UP = P$ for every $U \in \mathcal{S}$.
2. For every $f \in H$, the vector $Pf \in H$ is the unique element of $\overline{\text{co}}\{Uf \mid U \in \mathcal{S}\}$ of minimal norm.

Bergelson, V. (1985). 'Sets of recurrence of zm-actions and properties of sets of differences in zm', Journal of the London Mathematical Society, s2-31 (2), pp. 295–304. <https://doi.org/10.1112/jlms/s2-31.2.295>.

Lindenstrauss, E. (2001). 'Pointwise theorems for amenable groups', Inventiones mathematicae, 146 (2), pp. 259–295. <https://doi.org/10.1007/s002220100162>.

Host, B. (2019). 'A short proof of a conjecture of erdős proved by moreira, richter and robertson', Available at: <https://arxiv.org/abs/1904.09952>.

Jamneshan, A. and Kreidler, H. (2025). 'ISem 28: Ergodic structure theory and applications',.